Milestone 5
Velazquez-Feliciano, Adrian
DSC 640: Data Presentation and Visualization
03/08/2026

Final Submission:

The Childcare-Workforce Nexus: A Strategic Analysis of Economic Participation

Summary of Analysis

This project investigates the critical intersection of childcare affordability and the economic agency of families across the United States. Utilizing county-level indicators from the National Database of Childcare Prices (NDCP), the analysis explores how financial pressures related to childcare costs correlate with the Labor Force Participation Rate (LFPR) of mothers with children under age six. The research process began by inspecting variable structures within the NDCP, specifically identifying inflation-adjusted income measures and household composition variables to surface where affordability burdens are most acute.

The analysis is designed to be actionable rather than purely descriptive. It utilizes a strategic approach that combines geographic pattern-finding—to orient the audience toward regional clusters—with a relationship view that summarizes how affordability and workforce participation move in tandem.

Findings and Geographic Nuance

The primary finding across all visual mediums is a significant negative association between childcare affordability burden and maternal labor force participation. In regions where childcare consumes a larger share of median household income, mothers' participation in the labor force tends to be lower. To maintain analytical integrity, this relationship is framed as an association rather than direct causality.

A second critical finding is that this burden is not uniform. Significant state-to-state differences exist:

- Alabama: Families face a burden concentrated in lower-income counties where childcare price increases outpace wage growth.
- Florida: While statewide averages may appear more favorable, county-level data reveals clusters where high prices relative to income correlate with lower participation, particularly in high-cost metro corridors and seasonal economies.

Strategic Design and Mediums

Three distinct mediums were developed to match different decision-making contexts:

Interactive Dashboard: Supports exploration and allows users to identify specific "hotspots" through flexible filtering.

- *Executive Brief:* A clean, two-visual layout designed to move decision-makers from orientation to action in under three minutes.
- *Slide Narrative*: A modular presentation format intended for live discussion where trade-offs and next steps can be negotiated.
- *Design Decisions:* To reduce cognitive load, a single-hue red scale was utilized across all platforms; darker tones intuitively signal higher cost pressure. Consistent typography (Helvetica) and generous spacing ensure readability on laptops and projectors alike.

Ethical Considerations and Assumptions

The methodology relies on several key assumptions, including the use of standardized NDCP fields as comparable across time and the operationalization of the

as a valid proxy for family cost pressure.

$$affordability\ rate = \frac{Price\ * 12}{Median\ Household\ Income}$$

Ethical Mitigation:

- *Transparency*: All cleaning steps—including handling missing values and aligning time periods—are disclosed to maintain credibility.
- *Risk Management:* To avoid stigmatizing high-burden communities, the narrative emphasizes that these findings are intended to prioritize support and targeted action rather than implying causal failures within those regions.
- *Equity:* The inclusion of demographic fields allowed for equity-focused analyses, specifically comparing racial composition against affordability to highlight where interventions are most needed.

Persuasive Call to Action: Unlocking the Workforce

The central message of this project is that childcare is a fundamental component of economic infrastructure. To unlock local labor supply, a targeted, multi-stakeholder strategy must be implemented immediately:

- *For Policymakers & Workforce Boards:* Streamline subsidy utilization in high-burden counties with "one-stop" enrollment and proactive outreach to maximize existing resources.
- *For Employers & Chambers of Commerce:* Implement childcare benefits, flexible shifts, and backup care options to improve labor retention and reduce turnover costs.
- *For Community Organizations:* Expand the supply of care where it is missing, specifically targeting extended-hour care and micro-center licensing support to align with modern family schedules.

Lessons Learned

Reflecting on the project, earlier methodological documentation regarding missingness rules and year coverage would have ensured even tighter consistency across mediums. However, the most successful element was the integrated narrative: using the map to identify where the burden is highest, the scatter to explain how it relates to participation, and the brief to dictate what to do next.

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MEDIUM DELIVERY

Medium #1:

https://sites.google.com/view/childcaredashboard/home

The Dashboard I used a single-hue red scale to encode affordability burden so darker tones intuitively signal higher cost pressure, paired with clear legends and tooltips to keep cognitive load low.

Visual hierarchy favors the map full-width for immediate orientation, with compact side panels for filters and small multiples (if enabled) so users can compare states/counties without overcrowding.

Spacing and typography are kept generous and consistent—larger titles, medium axis labels, and tight gutters—to preserve readability on laptops and projectors while ensuring that interaction elements don't obscure the data.

Medium #2:

Executive Brief:
https://drive.google.com/file/d/1GDfTAIo6JbJunFqSeluuDT_oZ5J5I87C/view?usp=sharing

I chose a clean, two-visual layout to move decision-makers from orientation to action in under three minutes. The opening choropleth uses a single-hue red scale (Reds) to encode affordability burden with darker tones indicating higher burden. A single-hue scale minimizes cognitive load and communicates directionality (higher = worse) intuitively. The title and caption are short and declarative to clarify the message immediately.

The scatter plot uses a light grid and soft alpha (≈0.35) for points to avoid overplotting, with a thin black regression line to guide the eye to the overall relationship without implying causality. The x-axis label spells out the affordability formula to cut ambiguity. I kept y-axis labeling direct (Mothers' LFPR Under 6) and avoided abbreviations in the title. Margins are generous to prevent label collisions, and both visuals share consistent typography (Helvetica) with 1–2 font sizes larger for titles to support distance readability.

The Executive Brief, has spacing that follows a simple rhythm: title, figure, caption, then a slim methods box. The methods/notes section is concise and scannable with short lines and minimal jargon. Page order goes from national orientation (map) to analytic insight (scatter), which sets up the shortlist discussion in the slide deck and ensures the brief can stand alone as a leave-behind.

Medium #3:

Slide Deck:
https://docs.google.com/presentation/d/161pf57ZRm3HaVOTbpgp4MXED6XLKk1JN/edit?usp=sharing&ouid=116810142528822042760&rtpof=true&sd=true

The deck mirrors the brief's flow but optimizes for presentation. I use a white background and a restrained color palette centered on a deep red accent to align with the choropleth's semantics (darker red = higher burden). Typography is consistent (Helvetica) with clear hierarchy: large titles, medium labels, and small captions. Visuals occupy as much width as feasible to maintain readability in rooms—map and scatter are each full-width on their slides with a single-line caption below.

Bullets are capped at three to maintain pace and comprehension. The shortlist slide uses a tight two-column layout so criteria and counts are visible above the fold. Calls to action are audience-specific and structured in three columns so each stakeholder type sees themselves and their next step clearly.

Spacing follows a modular grid—consistent left margins, equal vertical spacing, and predictable caption placement—so viewers quickly parse slide structure as content changes. The optional state drill-down is placed after the national view to flex for different audiences without breaking the narrative arc. Finally, the resources/Q&A slides are intentionally minimal for discussion, with any tables (e.g., priority county list) moved to the appendix to avoid clutter.

All three visuals are independent, but work wonderfully together. Keep in mind that the slide deck could include a video recording for delivery .

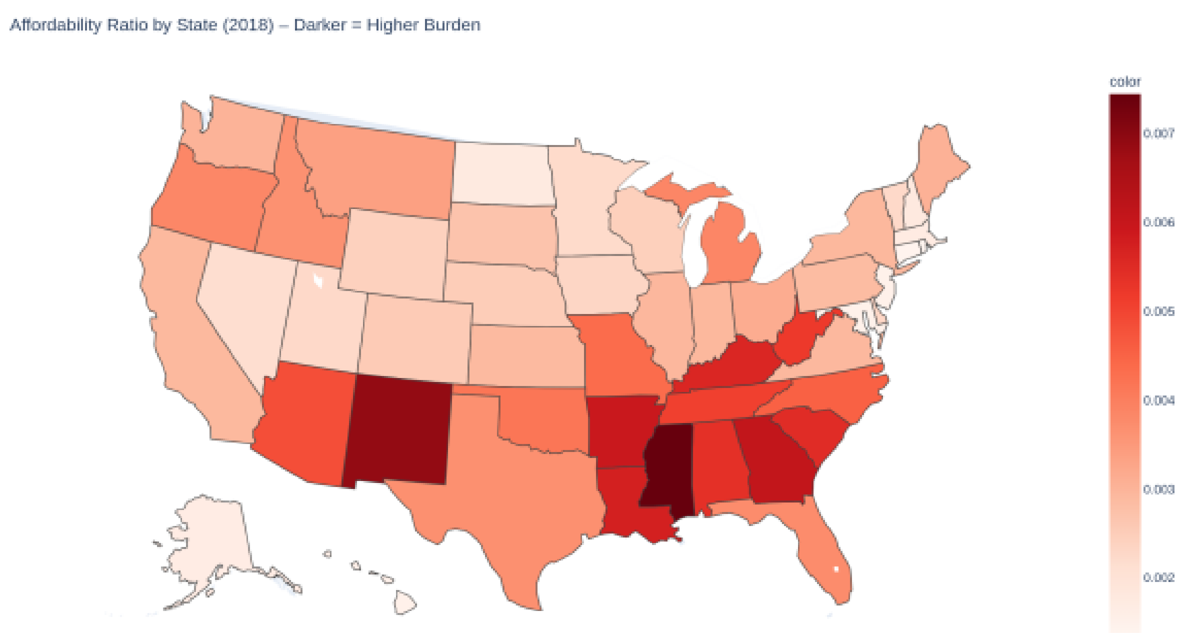
EXECUTIVE BRIEF: CHILDCARE AFFORDABILITY AND MOTHERS' LABOR FORCE PARTICIPATION

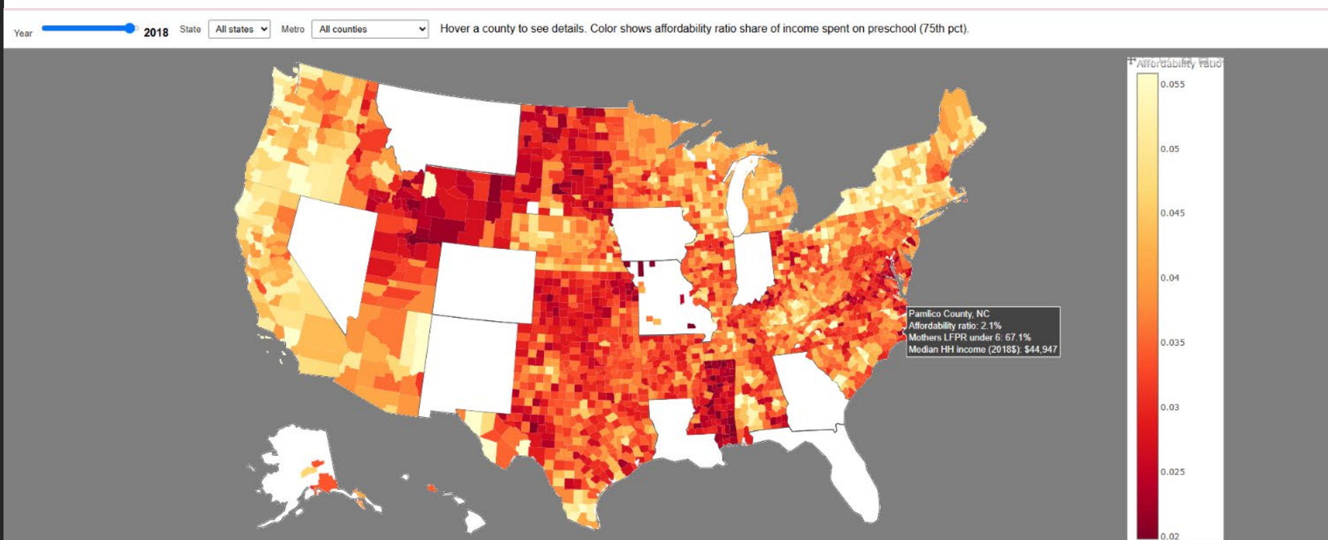
DSC-640

VELAZQUEZ, A.

Executive Brief: Childcare Affordability and Mothers LFPR

County-level insights for targeted action (2018 baseline)

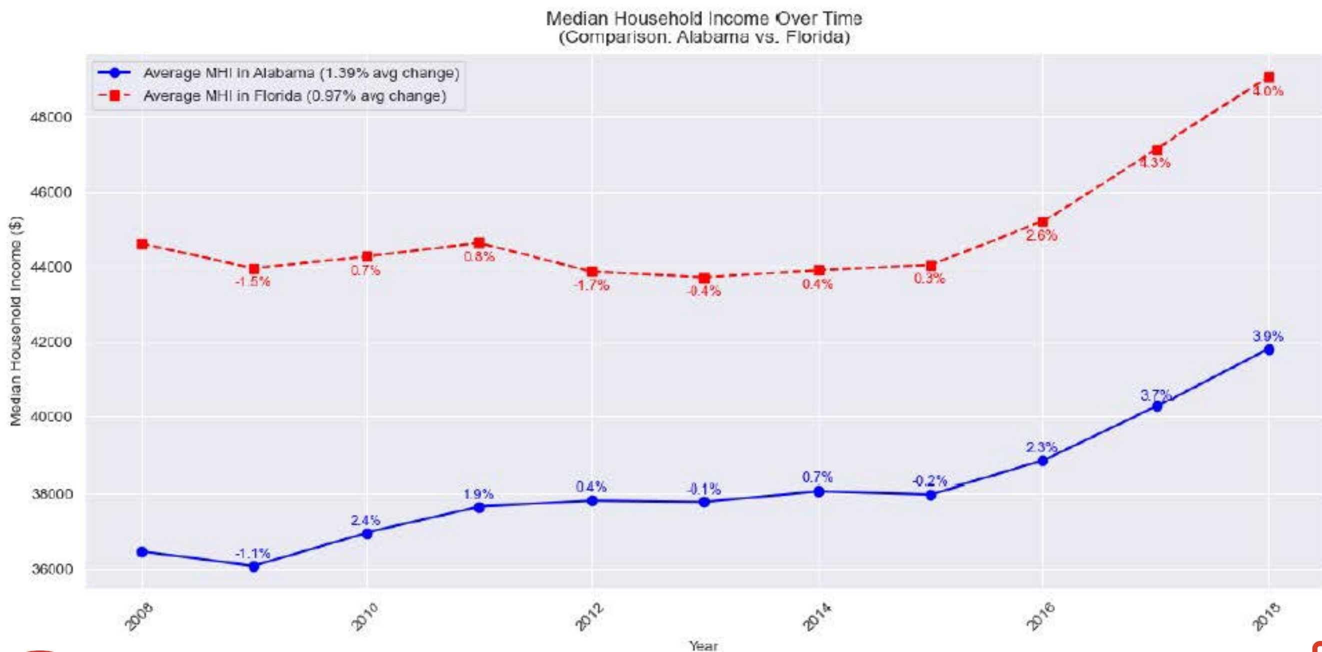




CHILDCARE AFFORDABILITY PRESSURES AND MOTHERS' LABOR FORCE PARTICIPATION (LFPR) CONVERGE TO SHAPE LOCAL LABOR SUPPLY. USING COUNTY-LEVEL DATA AND A 2018 BASELINE, WE SURFACE WHERE AFFORDABILITY BURDENS ARE HIGHEST AND HOW THEY RELATE TO MOTHERS' LFPR FOR CHILDREN UNDER AGE SIX. A STATE-LEVEL CHOROPLETH PROVIDES NATIONAL ORIENTATION, WHILE A COUNTY-LEVEL SCATTER WITH A REGRESSION LINE REVEALS A NEGATIVE ASSOCIATION BETWEEN AFFORDABILITY BURDEN AND MOTHERS' LFPR, HIGHLIGHTING PRIORITY COUNTIES WHERE TARGETED ACTIONS COULD UNLOCK LABOR SUPPLY.

Alabama and Florida face a shared challenge with different textures: families in high-burden counties spend a larger share of income on childcare, and mothers' labor force participation (LFPR) slips where affordability pressures bite hardest. In Alabama, the burden concentrates in lower-income counties where childcare prices outpace wage growth. In Florida, the picture is more mixed—higher incomes in metro corridors mask affordability hot spots in service- and tourism-heavy counties where schedules are irregular and non-traditional care is scarce. The takeaway is the same: affordability is suppressing participation, and that is suppressing local labor supply.

The 2018 baseline illustrates this clearly. Alabama's median affordability burden ranks among higher tiers in the Southeast, with pockets where mothers' LFPR under age six trails comparable counties. Florida's statewide averages look better at first glance, but county-level scatter points reveal clusters where high prices relative to income correlate with lower LFPR—especially in high-cost metros and seasonal economies. This is not theory: it is a map of lost shifts, unfilled roles, and foregone earnings for families.



CALL TO ACTION:	YOUR IDEA AND COMMITMENT
• Streamline Subsidy Utilization In High-Burden Counties With One-Stop Enrollment, Proactive Outreach, And Faster Reimbursements To Providers. • Expand Supply Where It Is Missing: Incentives For Extended-Hour Care, Micro-Center Licensing Support, And Shared Services For Small Providers. • Align Workforce Grants And Employer Partnerships To Alabama's And Florida's High-Burden/Low-Lfpr Clusters To Unlock Labor Supply Now..	Add notes, suggestion, commitment and feedback: ____________________ ____________________ ____________________ ____________________ ____________________ ____________________

CHILDCARE AFFORDABILITY AND MOTHERS’ LABOR FORCE PARTICIPATION

Title: Childcare Affordability and Mothers’ LFPR (2018 baseline)

DASHBOARD <- LINK

WHAT WE MEASURED: AFFORDABILITY RATIO: CALCULATED AS (PRICE X 12 / MEDIAN HOUSEHOLD INCOME) TO SHOW THE SHARE OF INCOME SPENT ON PRESCHOOL.

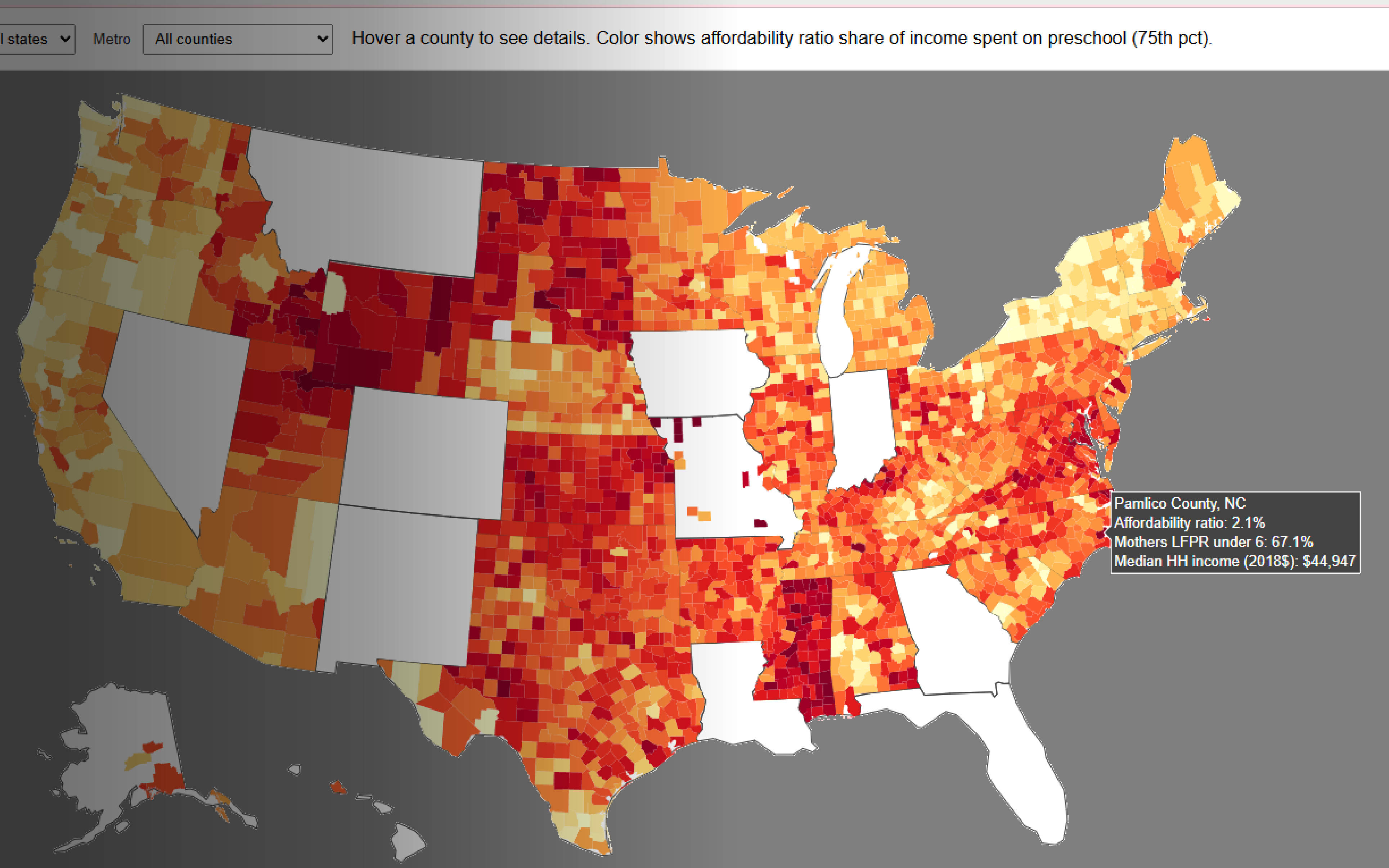
MOTHERS' LFPR: SPECIFICALLY TRACKS THE LABOR FORCE PARTICIPATION RATE FOR MOTHERS WITH CHILDREN UNDER AGE 6.

COVERAGE: COUNTY-LEVEL GRANULARITY ACROSS THE U.S. USING DATA FROM 2008–2018.

KEY INDICATORS: INCLUDES INFLATION-ADJUSTED HOUSEHOLD INCOME (MHI_2018) AND HOUSEHOLD COMPOSITION FOR CHILDREN UNDER 6.

NATIONAL MAP (2018) VISUAL AND OVERVIEW.

- CAPTION: DARKER = HIGHER AFFORDABILITY BURDEN



The Affordability-Participation Relationship

- **Visual Guide:**

  Use the national choropleth map where darker colors indicate a higher affordability burden.

- **Key Observation:**

  Higher burdens are prominently visible in regional clusters, particularly in parts of the South and Midwest.

- **Example Metric:**

  1. **In Pamlico County, NC, the affordability ratio was 2.1% with a Mothers' LFPR of 67.1% in 2018.**

Executive Brief: Childcare Affordability and Mothers LFPR

County-level insights for targeted action (2018 baseline)

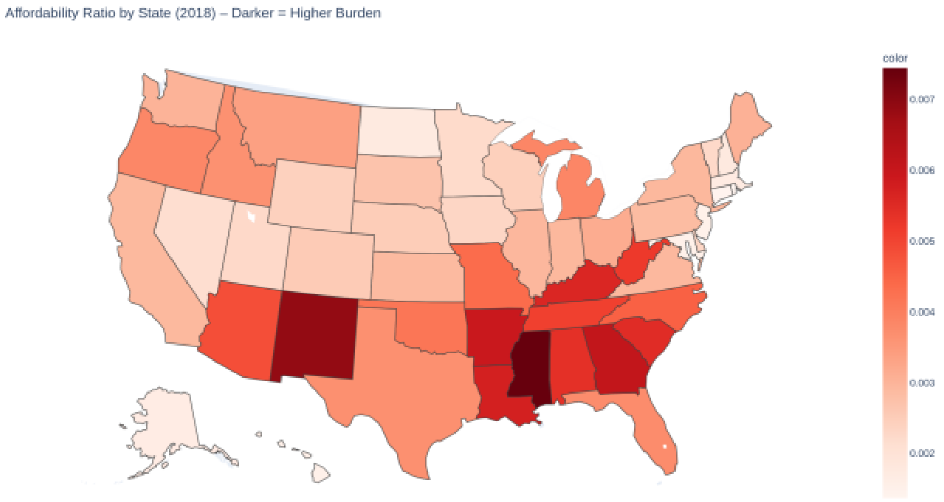
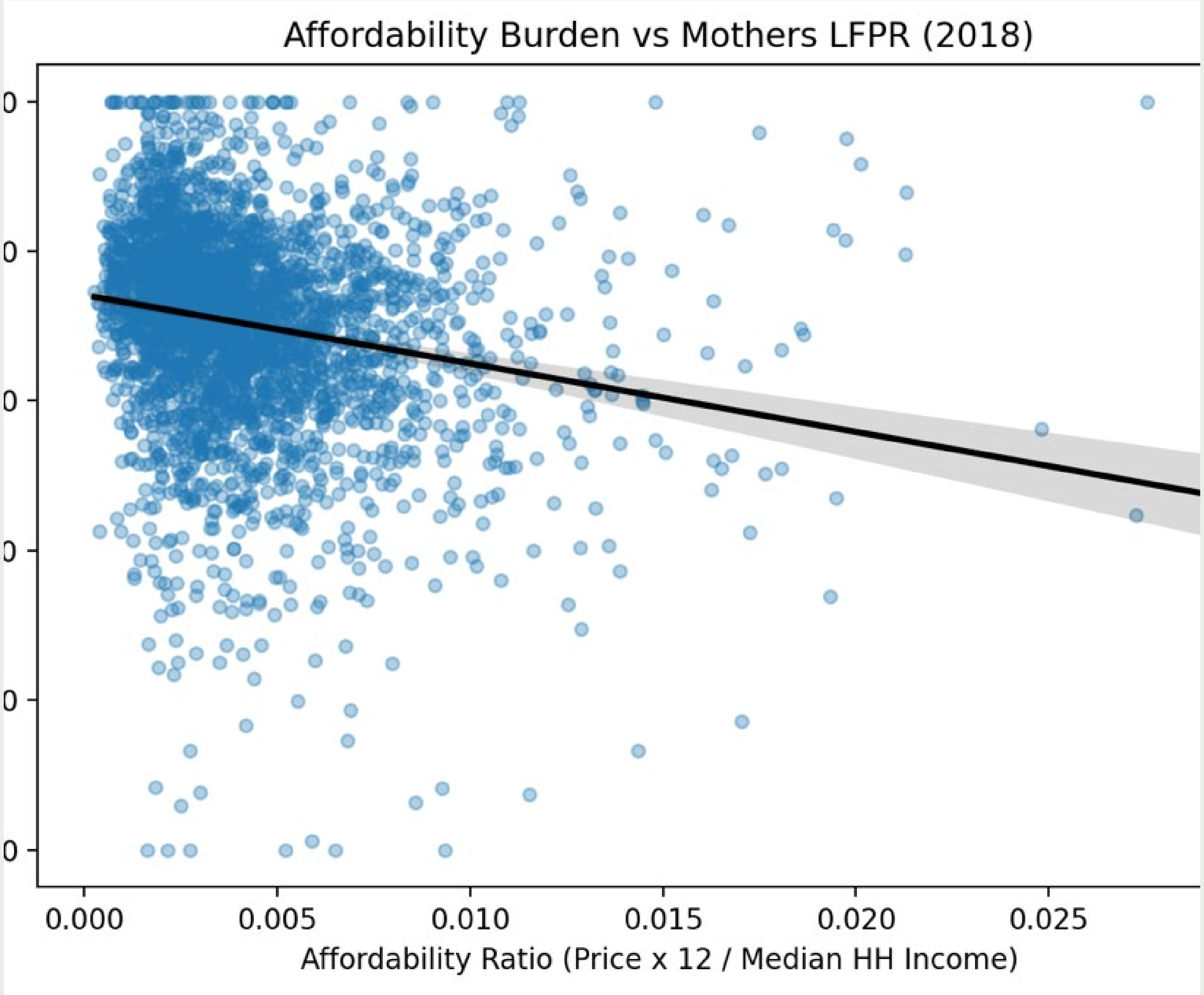


Figure 1. Affordability Ratio by State (2018). Darker = higher burden.

NATIONAL MAP (2018) VISUAL AND OVERVIEW.

- CAPTION: DARKER = HIGHER AFFORDABILITY BURDEN

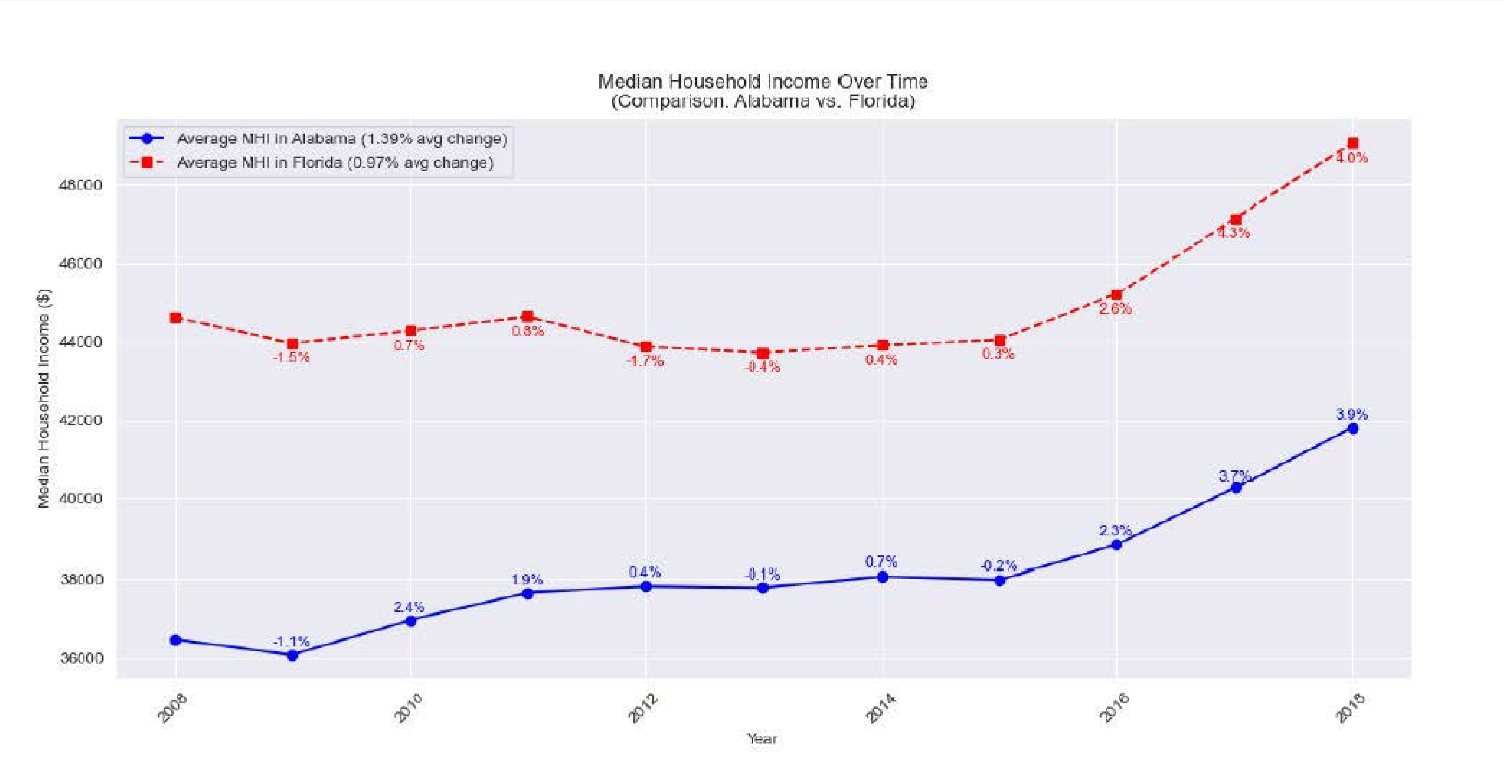


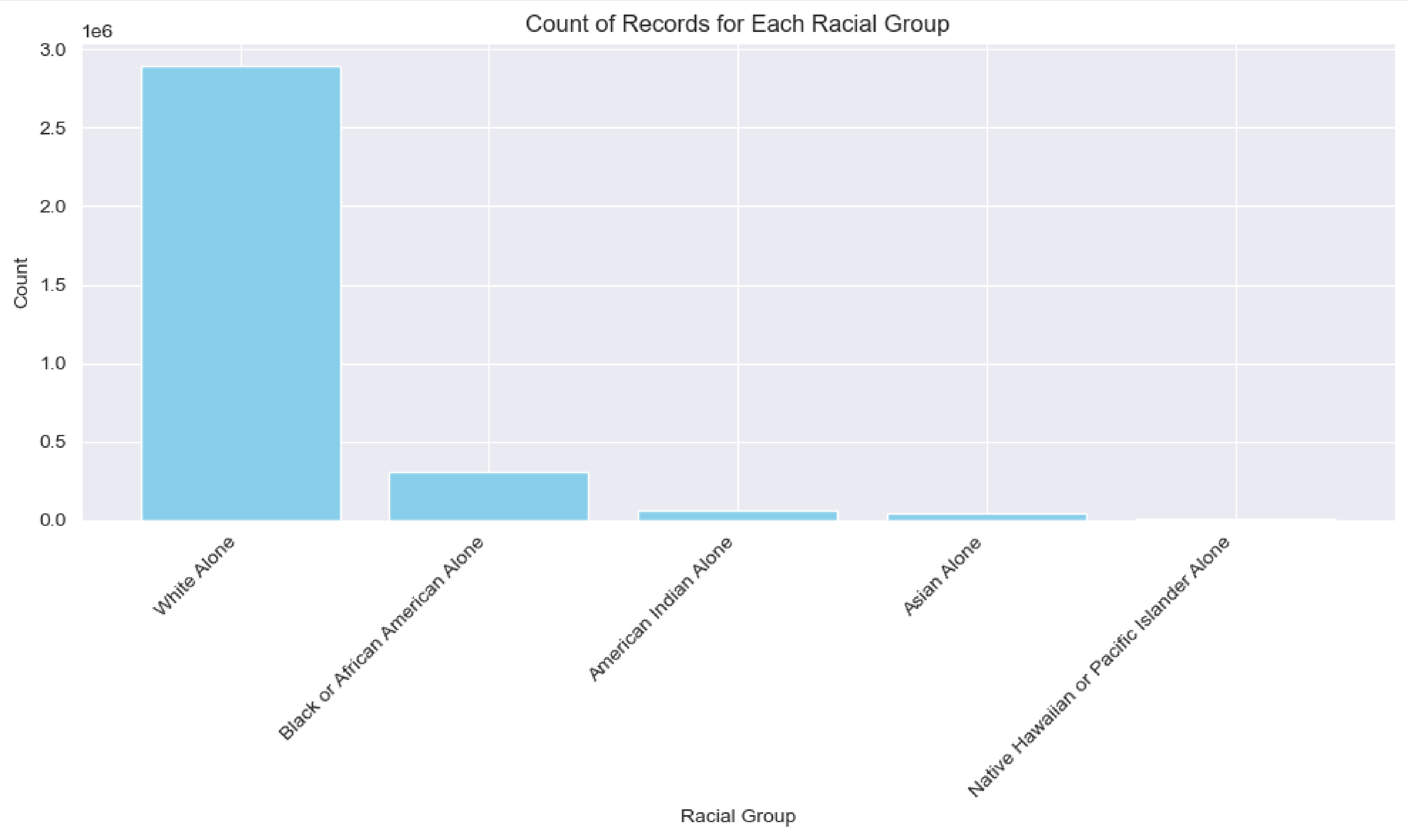
The Affordability-Participation Relationship

Core Finding:

There is a negative correlation where higher childcare burden often aligns with lower maternal labor force participation.

- **Target Quadrant:**
  Prioritize intervention for counties in the quadrant where high affordability ratios meet low LFPR.





Analytic Capability:

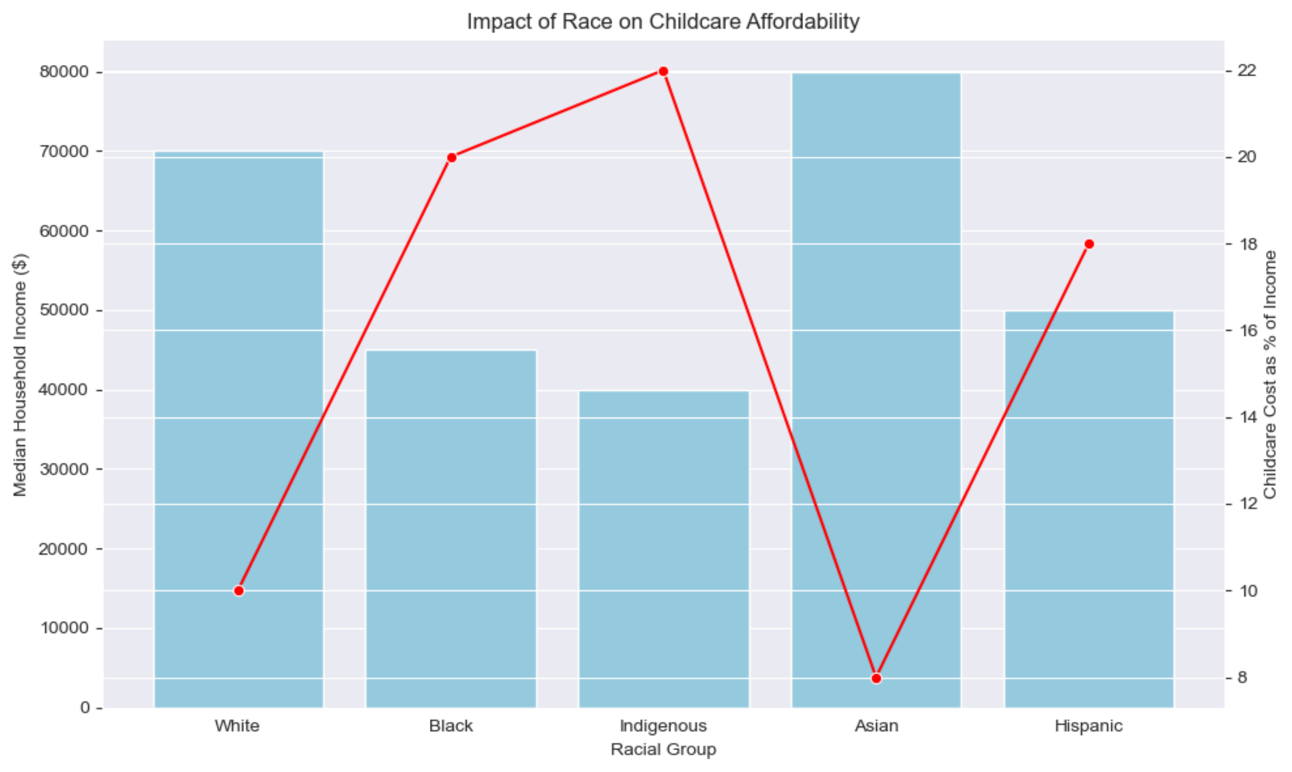
Considerations were given

Racial Group Distribution:
The dataset is predominantly composed of records identified as "White Alone"

Variables Included:

The data tracks demographic shares, including race/ethnicity and Hispanic identification

The inclusion of rich demographic fields allows for equity-focused analyses, specifically comparing racial composition against childcare affordability.



Calls to Action

Listen

Feedback

Commitment

Delivery

Audience – Calls to Actions

- Employers/Chambers: Implement childcare benefits, flexible shifts, and backup care options to improve labor retention.
- Policymakers/Workforce Boards: Focus on capacity matchmaking between providers and families and maximize subsidy utilization.
- Community Organizations: Conduct targeted outreach to families in high-burden tracts to connect them with available resources